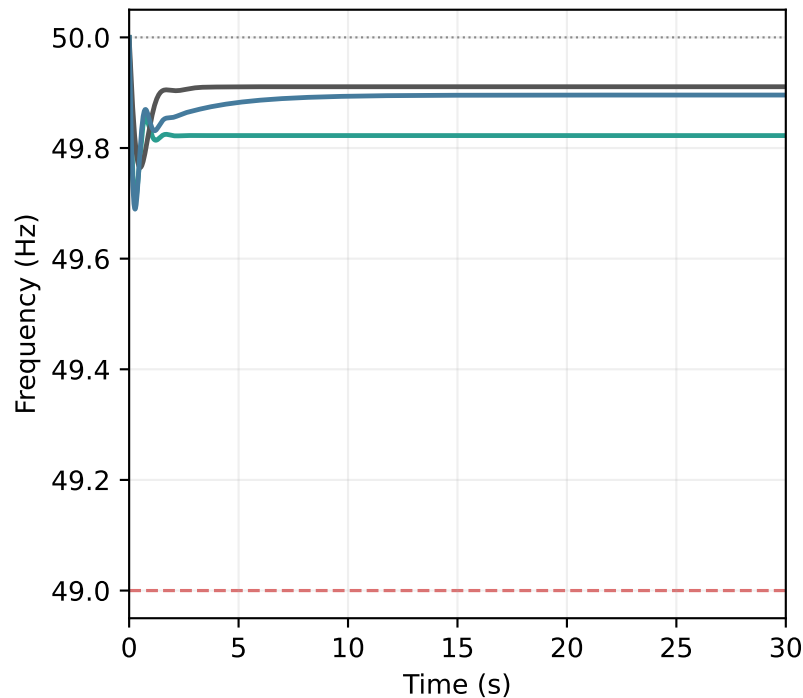


Primary Frequency Response Screening — Islanded Microgrid

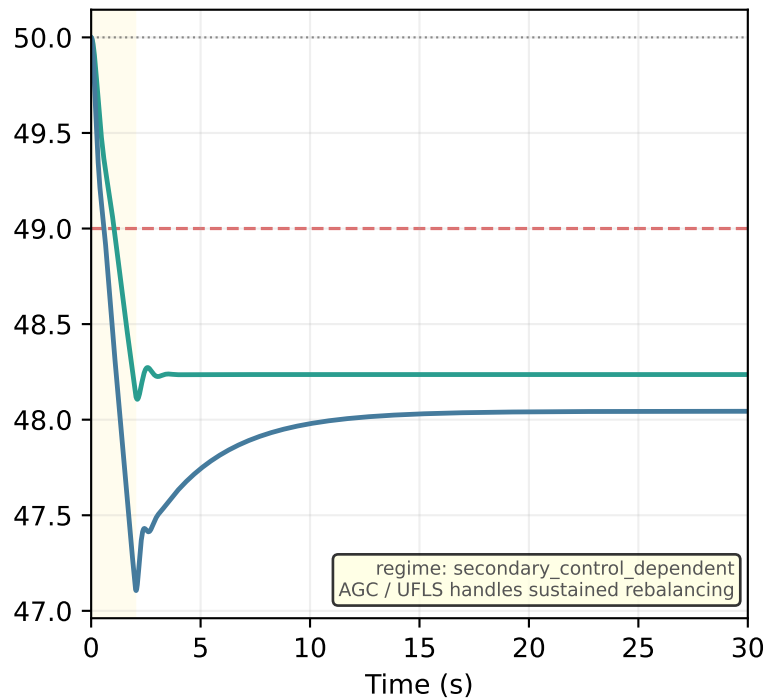
Primary response only. Sustained rebalancing after large PV loss or generator trip: secondary control / UFLS / dispatch (outside this model).

Panel A: primary_adequate (RoCoF + nadir assessed). Panels B-C: secondary_control_dependent (RoCoF only).

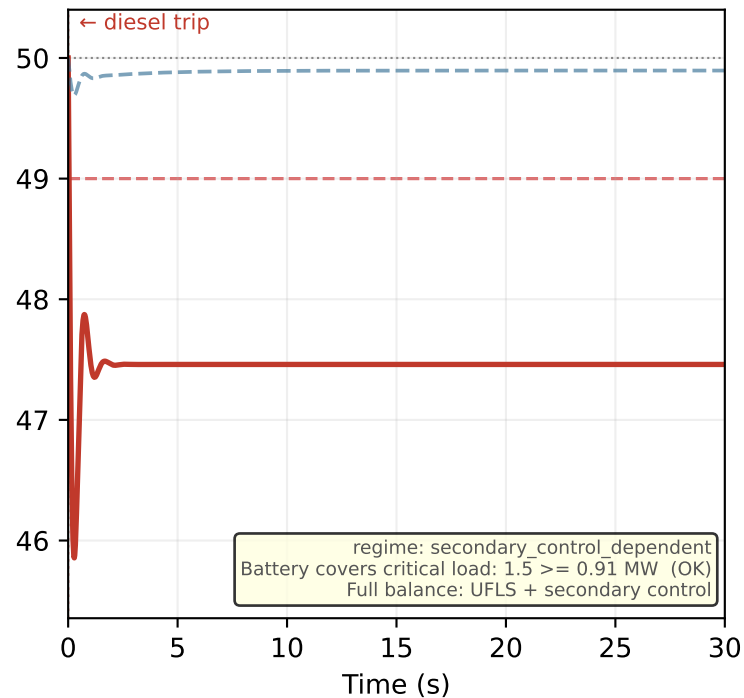
A 10 % Load Step — Evening peak
(all portfolios, primary response only)



B 50 % PV Ramp — Solar period
(battery-dominated portfolios)



C Diesel Trip + Grid-Forming Handover
(diesel_battery: battery takes over GF at t=0)



PV+battery (bat GF) [$H_{eq}=1.33s$]

PV+bat+H₂ (bat GF) [$H_{eq}=1.21s$]

PV+battery (bat GF) (PV=3.21 MW)

PV ramp (2s)

PV+diesel+bat (bat GF, post-trip)
 $\Delta P=1.65$ MW, $H_{eq}=0.91s$

49.0 Hz threshold

PV+diesel+bat (diesel GF) [$H_{eq}=2.91s$]

49.0 Hz threshold

PV+bat+H₂ (bat GF) (PV=8.08 MW)

49.0 Hz threshold

H₂+battery load step (ref)